電子學實驗 (一)
Integrators, Differentiators, and Their Frequency Responses
結報

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1 實驗結果、Bode Plot 與數據分析

I Exp. 1 Integrator

i Circuit

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input resistance $R = 1$ kΩ(current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a capacitance $C = 0.01$ μF (103) (current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

ii Theoretical results

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Time domain:

$$v_+ = v_- = 0$$

$$i_I = \frac{v_I}{R} = i_C = -C\frac{\mathrm{d}v_O}{\mathrm{d}t}$$

$$v_O = v_O(0^-) - \frac{1}{RC}\int_0^t v_I(\tau)\,\mathrm{d}\tau$$

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_C = -CsV_O + Cv_O(0^-)$$

$$V_O = \frac{1}{s}v_O(0^-) - \frac{1}{RCs}V_I$$

Transfer function: Infinite DC gain. 90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -\frac{1}{RCs}$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Zero-state responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

$$v_O(t) = -\frac{AT}{RC}\left(2s\,|s| - s\right), \quad s = \frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

Triangle wave output with amplitude $\frac{AT}{4RC}$

$$v_O(t) = -\frac{AT}{RC}\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right)$$

- Sinusoidal input:

$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

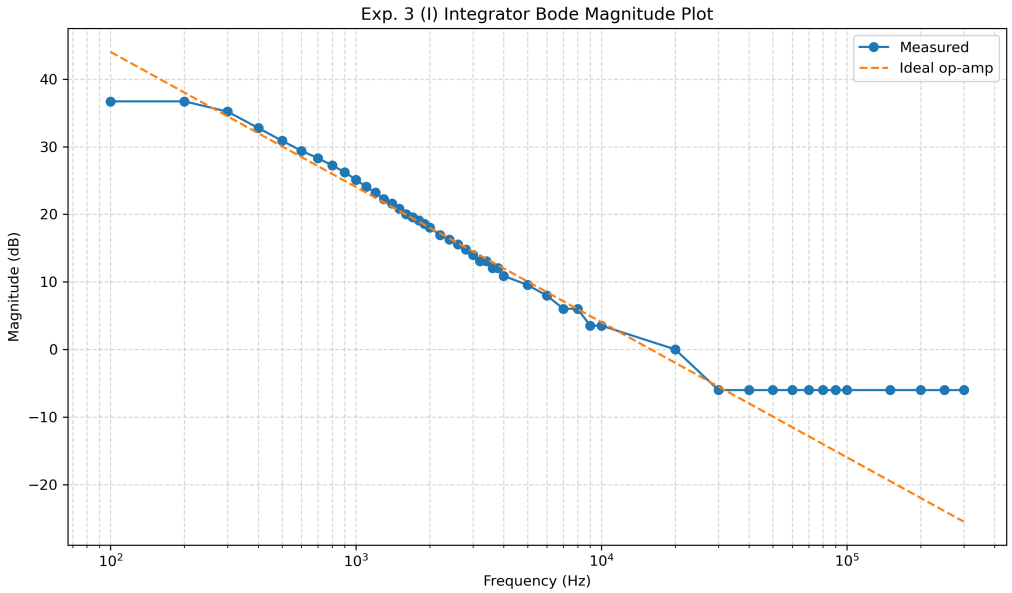
Sinusoidal output 90 degrees leading input with amplitude $\frac{AT}{RC}$.

$$v_O(t) = \frac{AT}{RC}\cos\left(\frac{t}{T}\right)$$

iii Measured data

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
100	27.40	36.71	63.66	44.04	-56.96
200	27.40	36.71	31.83	38.02	-13.92
300	23.00	35.19	21.22	34.49	8.38
400	17.40	32.77	15.92	32.00	9.33
500	14.00	30.88	12.73	30.06	9.96
600	11.80	29.40	10.61	28.47	11.21
700	10.40	28.30	9.09	27.13	14.35
800	9.20	27.23	7.96	25.97	15.61
900	8.20	26.24	7.07	24.95	15.92
1000	7.20	25.11	6.37	24.04	13.10
1100	6.40	24.08	5.79	23.21	10.58
1200	5.80	23.23	5.31	22.45	9.33
1300	5.20	22.28	4.90	21.76	6.19
1400	4.80	21.58	4.55	21.11	5.56
1500	4.40	20.83	4.24	20.51	3.67
1600	4.00	20.00	3.98	19.95	0.53
1700	3.80	19.55	3.74	19.43	1.47
1800	3.60	19.08	3.54	18.93	1.79
1900	3.40	18.59	3.35	18.46	1.47
2000	3.20	18.06	3.18	18.02	0.53
2200	2.80	16.90	2.89	17.19	-3.24
2400	2.60	16.26	2.65	16.43	-1.98
2600	2.40	15.56	2.45	15.74	-1.98
2800	2.20	14.81	2.27	15.09	-3.24
3000	2.00	13.98	2.12	14.49	-5.75
3200	1.80	13.06	1.99	13.93	-9.52
3400	1.80	13.06	1.87	13.41	-3.87
3600	1.60	12.04	1.77	12.91	-9.52
3800	1.60	12.04	1.68	12.44	-4.50
4000	1.40	10.88	1.59	12.00	-12.04
5000	1.20	9.54	1.27	10.06	-5.75
6000	1.00	7.96	1.06	8.47	-5.75
7000	0.80	6.02	0.91	7.13	-12.04
8000	0.80	6.02	0.80	5.97	0.53
9000	0.60	3.52	0.71	4.95	-15.18
10000	0.60	3.52	0.64	4.04	-5.75
20000	0.40	0.00	0.32	-1.98	25.66
30000	0.20	-6.02	0.21	-5.51	-5.75
40000	0.20	-6.02	0.16	-8.00	25.66
50000	0.20	-6.02	0.13	-9.94	57.08
60000	0.20	-6.02	0.11	-11.53	88.50
70000	0.20	-6.02	0.09	-12.87	119.91
80000	0.20	-6.02	0.08	-14.03	151.33
90000	0.20	-6.02	0.07	-15.05	182.74
100000	0.20	-6.02	0.06	-15.96	214.16
150000	0.20	-6.02	0.04	-19.49	371.24
200000	0.20	-6.02	0.03	-21.98	528.32
250000	0.20	-6.02	0.03	-23.92	685.40
300000	0.20	-6.02	0.02	-25.51	842.48

iv Bode plot



II Exp. 2 Differentiator

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input capacitor $C = 0.01$ µF (103) (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a resistor $R = 1$ kΩ(current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

Theoretical results:

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Time domain:

$$i_I = C\frac{\mathrm{d}v_I}{\mathrm{d}t} = i_f = \frac{-v_O}{R}$$

$$v_O = -RC\frac{\mathrm{d}v_I}{\mathrm{d}t}$$

Laplace domain:

$$I_I = CsV_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -RCsV_I + RCv_I(0^-)$$

Transfer function: Infinite gain at $\omega = \infty$. -90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -RCs$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

  Square wave with amplitude $\frac{4ARC}{T}$

$$v_O(t) = -\frac{8ARC}{T}\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad \frac{2t}{T} \notin \mathbb{Z}$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Dirac delta sum output

$$v_O(t) = 2ARC\sum_{n\in\mathbb{Z}} \delta(t - nT) - \delta(t - \left(n + \frac{1}{2}\right)T)$$

- Sinusoidal input:

$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

  Sinusoidal output 90 degrees lagging input with amplitude $\frac{ARC}{T}$.

$$v_O(t) = -\frac{ARC}{T}\cos\left(\frac{t}{T}\right)$$

III Exp. 3 Frequency Responses (I)

Measure frequency response of the circuit in Exp. 1 Integrator under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 300\text{ kHz}$$

and sketch the Bode magnitude plot.

Measure frequency response of the circuit in Exp. 2 Differentiator under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 100\text{ kHz}$$

and sketch the Bode magnitude plot.

IV Exp. 3 Frequency Responses (II)

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 1\text{ MHz}$$

and sketch the Bode magnitude plot.

Theoretical results:

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 159 \text{ kHz}$$

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) and a resistor R_F in parallel connection under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

Theoretical results:

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_F = -\left(\frac{1}{R_F} + Cs\right)V_O + Cv_O(0^-) = -\frac{1 + R_F Cs}{R_F}V_O + Cv_O(0^-)$$

$$V_O = \frac{R_F C}{1 + R_F Cs}v_O(0^-) - \frac{\frac{R_F}{R}}{1 + R_F Cs}V_I$$

Transfer function: STC low-pass with cutoff frequency $\frac{1}{2\pi R_F C}$ and DC gain $\frac{R_F}{R}$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R_F}{R}}{1 + R_F Cs}$$

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